



CS215 DISCRETE MATH

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Cryptography

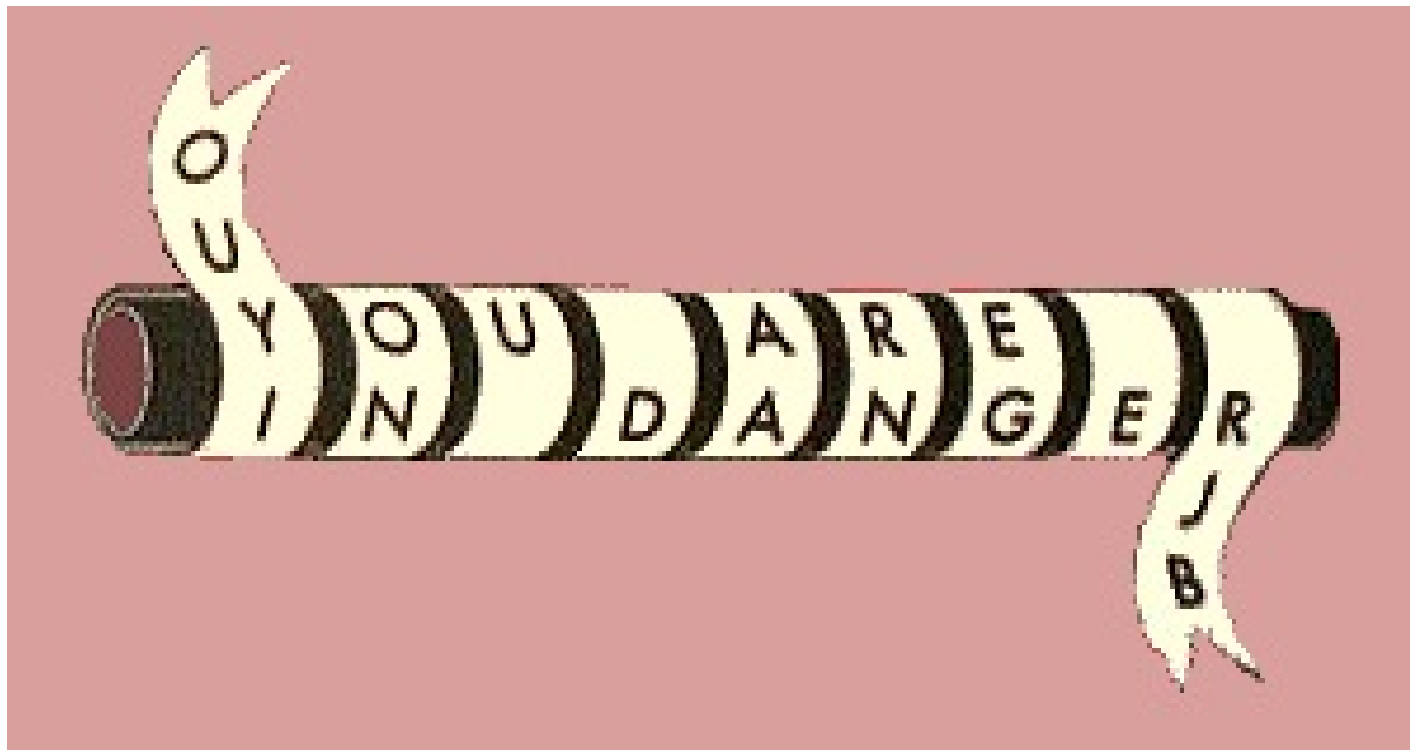
- One-sentence definition:

“Cryptography is the practice and study of techniques for secure communication in the presence of third parties called *adversaries*.” – Ronald L. Rivest



Some Examples

- In 405 BC, the Greek general LYSANDER OF SPARTA was sent a coded message written on the inside of a servant's belt.



Some Examples

- The Greeks also invented a cipher which changed **letters** to **numbers**. A form of this code was still being used during *World War I*.

	1	2	3	4	5
1	A	B	C	D	E
2	F	G	H	I/J	K
3	L	M	N	O	P
4	Q	R	S	T	U
5	V	W	X	Y	Z

Some Examples

- Caesar Cipher (after the name of JULIUS CAESAR)



VENI, VIDI, VICI

YHQL YLGL YLFL

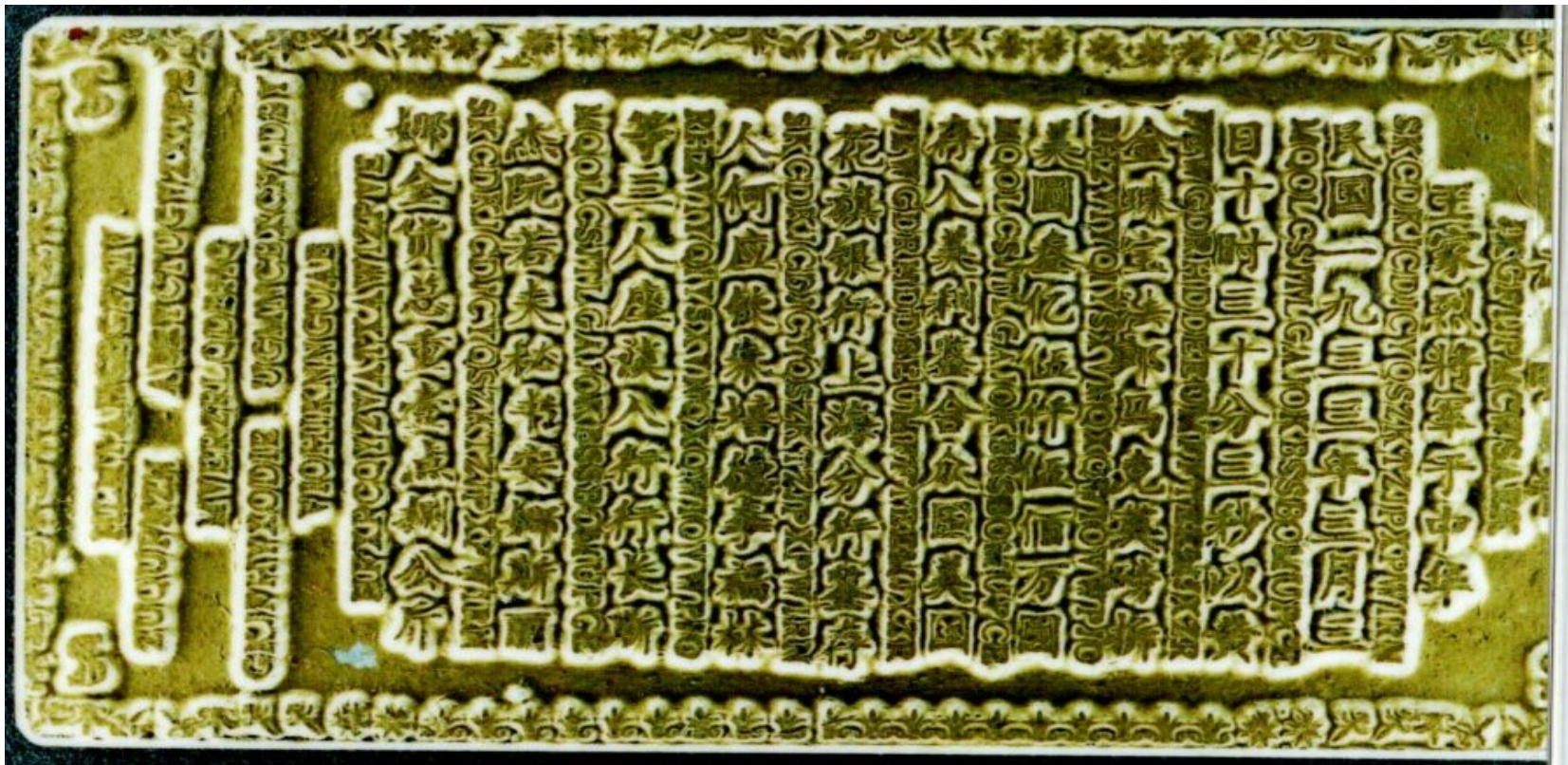
Some Examples

- Morse Code: created by Samuel Morse in 1838

Morse Alphabet	
A • —	O — — —
B — • • •	P • — — •
C — • — •	Q — — • —
D — • •	R • — •
E •	S • • •
F • • — •	T —
G — — •	U • • —
H • • • •	V • • • —
I • •	W • — —
J • — — —	X — • • —
K — • —	Y — • — —
L • — • •	Z — — • •
M — —	
N — •	
Full stop (.) • — • — • —	
Break signal or fresh line — • • • —	
Apostrophe (') • — — — — •	
Hyphen (-) — • • • • —	
Exclamation (!) — — • • — —	
Interrogation (?) • • — — • •	
Underline (———) • • — — • —	
Parenthesis () — • — — • —	
Inverted commas (" ") • — • • — •	

Some Examples

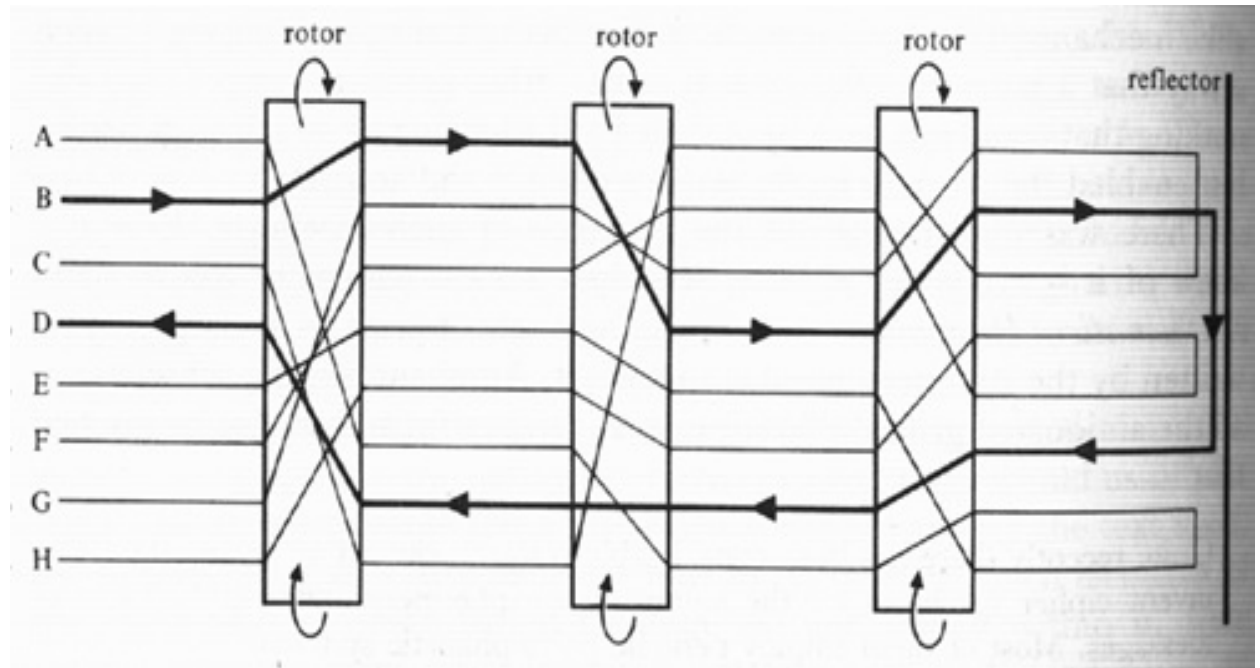
- Cryptograms from the Chinese gold bars



<http://www.iacr.org/misc/china/china.html>

Some Examples

- Enigma, Germany coding machine in *World War II*.



Some Examples

- Sigaba, used by U.S. during *World War II*.



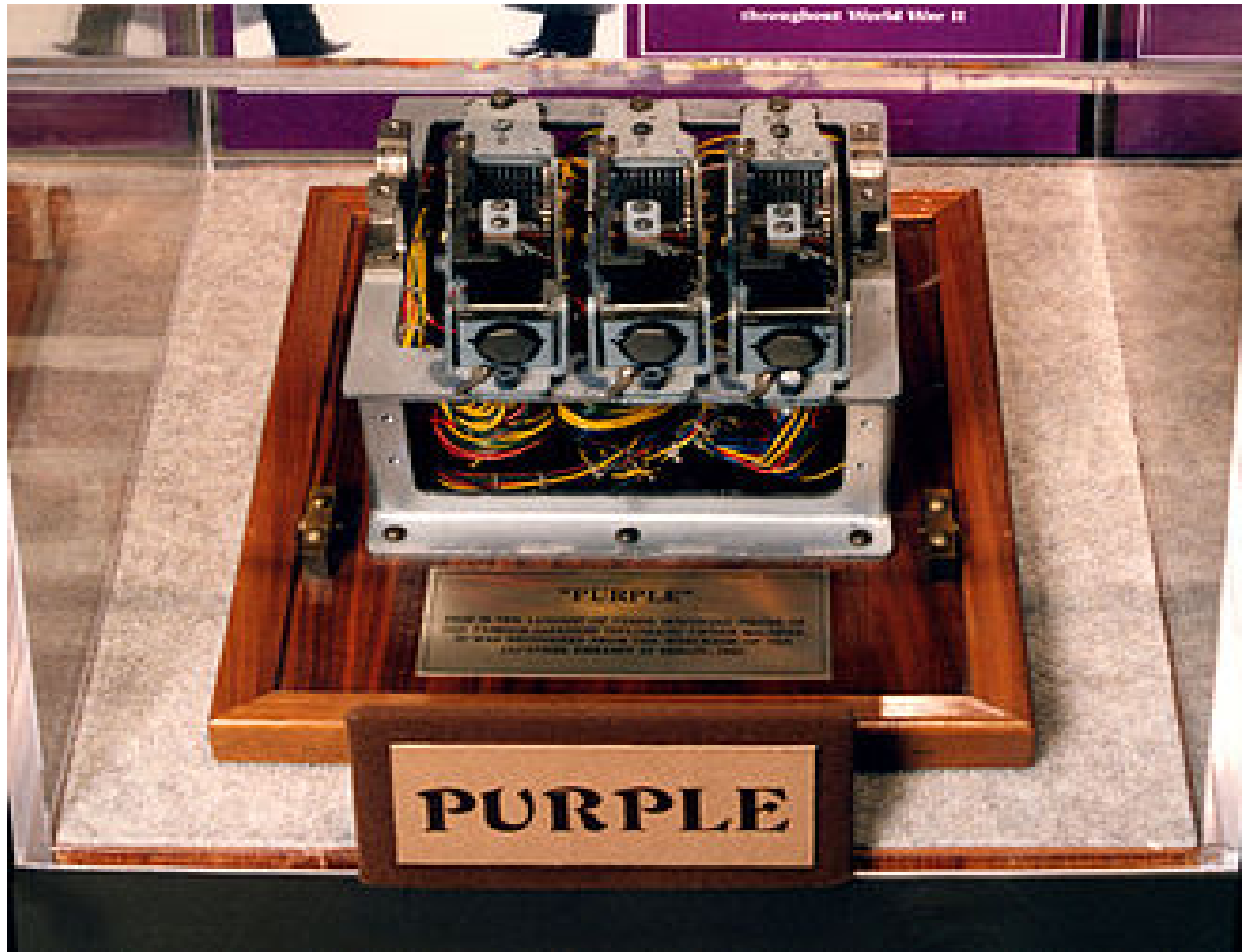
Some Examples

- Japanese “Enigma” Rotor Cipher Machine



Some Examples

- Japanese Purple Machine (97-shiki obun inji-ki)



People Working in Breaking Codes



Alan Turing
(1912-1954)



Claude E. Shannon
(1916-2001)

Cryptography History

- History (until 1970's)
 - “*Symmetric*” cryptography



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They need agree in advance on the *secret key* k .

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They need agree in advance on the **secret key k** .

Q: How can they do this?

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“*Symmetric*” cryptography



They need agree in advance on the **secret key k** .

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Q: What if Bob could send Alice a “special key” useful only for **encryption** but no help for **decryption**?

Caesar Cipher

■ **Key:** $k = 0, 1, \dots, 25$

Encryption: encode i as $(i + k) \bmod 26$

Decryption: decode j as $(j - k) \bmod 26$



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plaintext: SEND REINFORCEMENT

Key: 2

ciphertext: UGPF TGKPHQTEGOGPV



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Problem: only 26 possibilities for keys!

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Kerchoff's Principle (1883): System should be secure even if algorithms are known, as long as key is secret.



Substitution Cipher

- **Key:** table mapping each letter to another letter

A	B	C		Z
V	R	E		D



Substitution Cipher

- **Key:** table mapping each letter to another letter

A	B	C		Z
V	R	E		D

Encryption & Decryption: letter by letter according to table

of possible keys: $26! \approx 4 \times 10^{26}$



Substitution Cipher

■ Table 1: Relative frequencies of the letters of the English language

Letter	Relative Frequency (%)	Letter	Relative Frequency (%)
a	8.167	n	6.749
b	1.492	o	7.507
c	2.782	p	1.929
d	4.253	q	0.095
e	12.702	r	5.987
f	2.228	s	6.327
g	2.015	t	9.056
h	6.094	u	2.758
i	6.966	v	0.978
j	0.153	w	2.360
k	0.772	x	0.150
l	4.025	y	1.974
m	2.406	z	0.074

Substitution Cipher

Table 2: Number of Digraphs Expected in 2,000 Letters of English Text

th	-	50	at	-	25	st	-	20
er	-	40	en	-	25	io	-	18
on	-	39	es	-	25	le	-	18
an	-	38	of	-	25	is	-	17
re	-	36	or	-	25	ou	-	17
he	-	33	nt	-	24	ar	-	16
in	-	31	ea	-	22	as	-	16
ed	-	30	ti	-	22	de	-	16
ne	-	30	to	-	22	rt	-	16
ha	-	26	it	-	20	ve	-	16

Table 3: The 15 Most Common Trigraphs in the English Language

1	-	the	6	-	tio	11	-	edt
2	-	and	7	-	for	12	-	tis
3	-	tha	8	-	nde	13	-	oft
4	-	ent	9	-	has	14	-	sth
5	-	ion	10	-	nce	15	-	men

Substitution Cipher

- LIVITCSWPIYVEWHEVSRIQMXLEYVEOIEWHRXEXIPFE
MVEWHKVESTYLXZIXLIKIIXPIJVSZEYPERRGERIMWQL
MGLMXQERIWGPSRIHMXQEREKI



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I – *most common letter*

LI – *most common pair*

XLI – *most common triple*

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I = e

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L = h

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Y = g

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V = r

E = a

Y = g

HereUpOnLeGrandAroseWithAGraveAndStatelyAirAndBroug
MeTheBeetleFromAGlassCaseInWhichItWasEnclosedIt-
WasABe

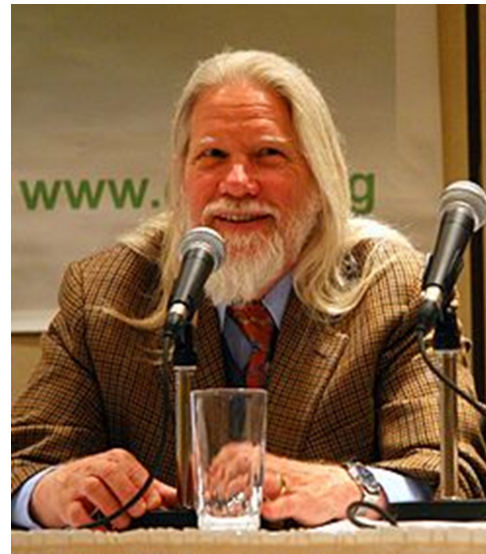


Cryptography History

- History (from 1976)

- ◇ W. Diffie, M. Hellman, “New direction in cryptography”, *IEEE Transactions on Information Theory*, vol. 22, pp. 644-654, 1976.

“We stand today on the brink of a revolution in cryptography.”



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2015 **Turing Award**



Bailey W. Diffie



Martin E. Hellman

2015	Martin E. Hellman Whitfield Diffie	For fundamental contributions to modern cryptography . Diffie and Hellman's groundbreaking 1976 paper, "New Directions in Cryptography," ^[39] introduced the ideas of public-key cryptography and digital signatures, which are the foundation for most regularly-used security protocols on the internet today. ^[40]
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Public Key Cryptography

- Alice wants to send a message to Bob



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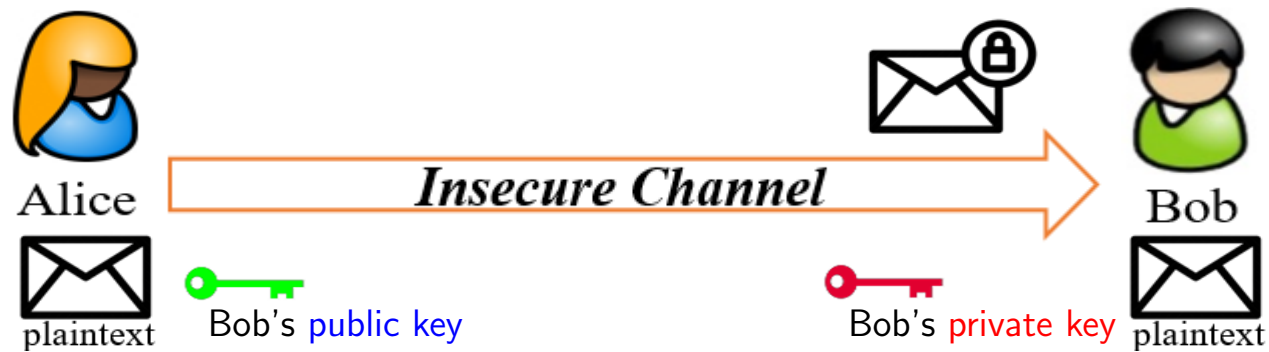
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Ronald L. Rivest



Adi Shamir



Leonard M. Adleman

R. Rivest, A. Shamir, L. Adleman, "A method for obtaining digital signatures and public-key cryptosystems",

Communications of the ACM, vol. 21-2, pages 120-126, 1978.

RSA Public Key Cryptosystem

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Pick two **large** primes, p and q . Let $n = pq$, then $\phi(n) = (p - 1)(q - 1)$. Encryption and decryption keys e and d are selected such that

- $\gcd(e, \phi(n)) = 1$
- $ed \equiv 1 \pmod{\phi(n)}$



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$$M = C^d \bmod n \text{ (RSA decryption)}$$



RSA Public Key Cryptosystem

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$$M = C^d \bmod n \text{ (RSA **decryption**)}$$

Theorem (*Correctness*) : Let p and q be two odd primes, and define $n = pq$. Let e be relatively prime to $\phi(n)$ and let d be the multiplicative inverse of e modulo $\phi(n)$. For each integer x such that $0 \leq x < n$,

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Q : How to prove this?



RSA Public Key Cryptosystem: Example

Parameters:	p	q	n	$\phi(n)$	e	d
	5	11	55	40	7	23



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Parameters:

p	q	n	$\phi(n)$	e	d
5	11	55	40	7	23

Public key: (7, 55)

Private key: 23

Encryption: $M = 28, C = M^7 \bmod 55 = 52$

Decryption: $M = C^{23} \bmod 55 = 28$



RSA Public Key Cryptosystem: Parameters

Parameters: p q n $\phi(n)$ e d

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$p, q, \phi(n)$ must be kept **secret**!



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Q : Why?

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Comment: It is believed that determining $\phi(n)$ is **equivalent** to factoring n . Meanwhile, determining d given e and n , appears to be at least as time-consuming as **the integer factoring problem**.



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CS 208 – Algorithm Design and Analysis



The Security of the RSA

In practice, RSA keys are typically 1024 to 2048 bits long.



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Q : Consider the RSA system, where $n = pq$ is the modulus. Let (e, d) be a key pair for the RSA. Define

$$\lambda(n) = \text{lcm}(p - 1, q - 1)$$

and compute $d' = e^{-1} \bmod \lambda(n)$. Will decryption using d' instead of d still work?



Applications of RSA

- SSL/TLS protocol



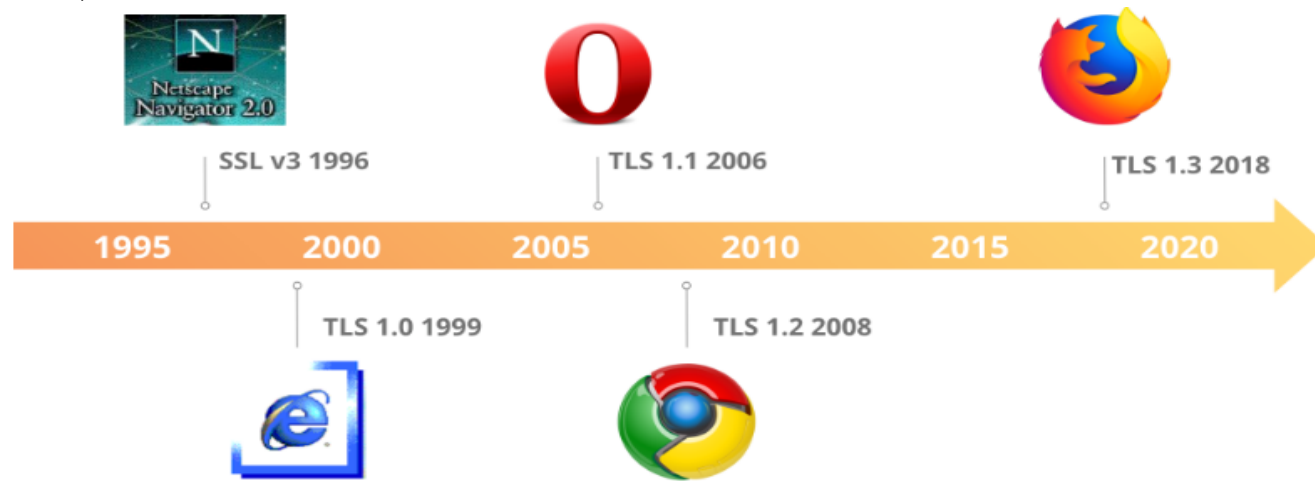
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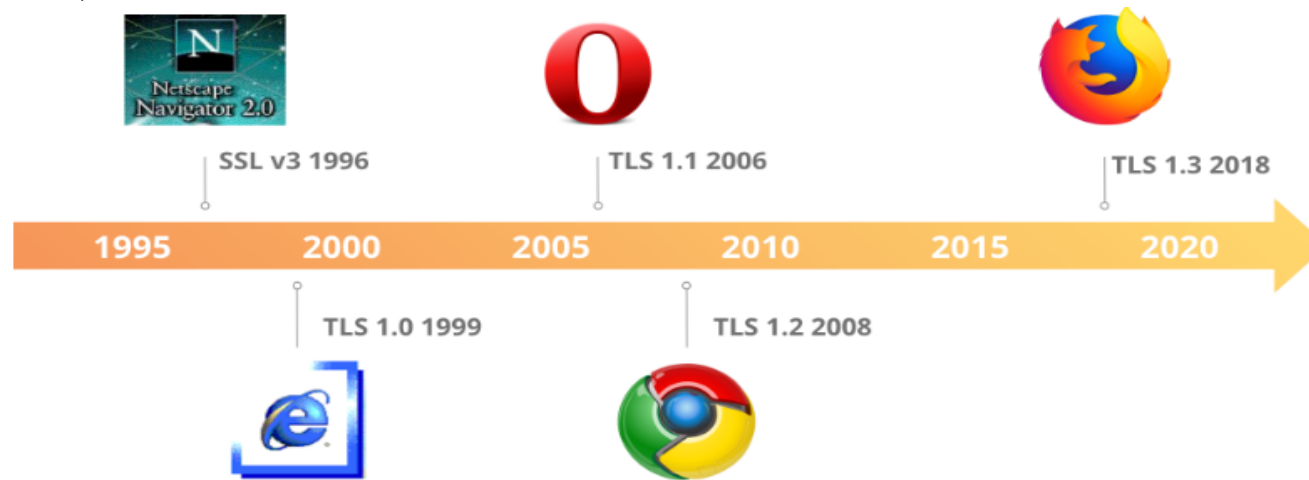
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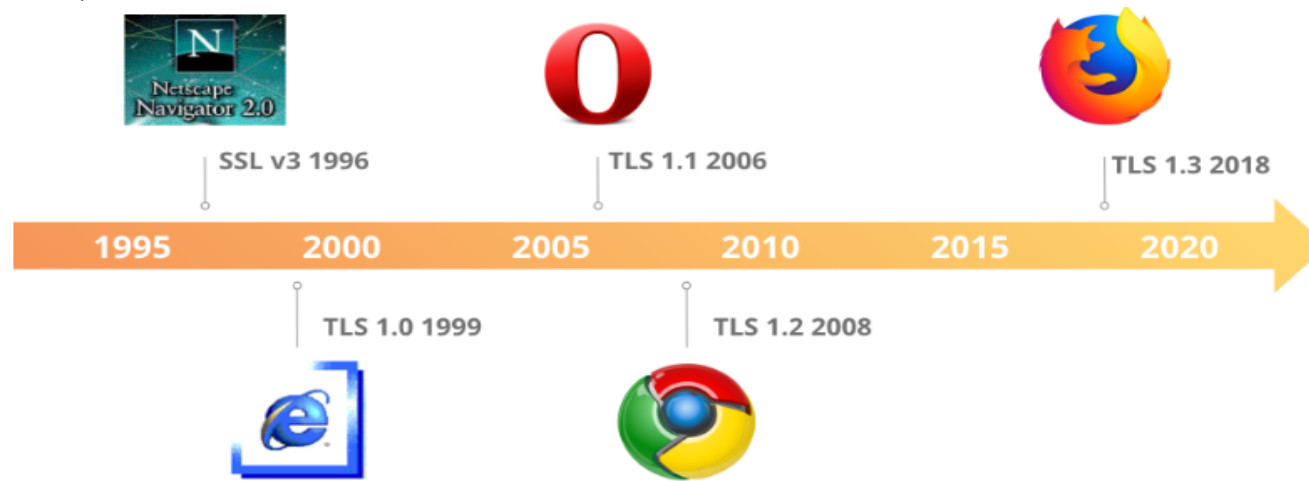


Key exchange/agreement and authentication

Algorithm	SSL 2.0	SSL 3.0	TLS 1.0	TLS 1.1	TLS 1.2	TLS 1.3
RSA	Yes	Yes	Yes	Yes	Yes	No
DH-RSA	No	Yes	Yes	Yes	Yes	No
DHE-RSA (forward secrecy)	No	Yes	Yes	Yes	Yes	Yes
ECDH-RSA	No	No	Yes	Yes	Yes	No
ECDHE-RSA (forward secrecy)	No	No	Yes	Yes	Yes	Yes

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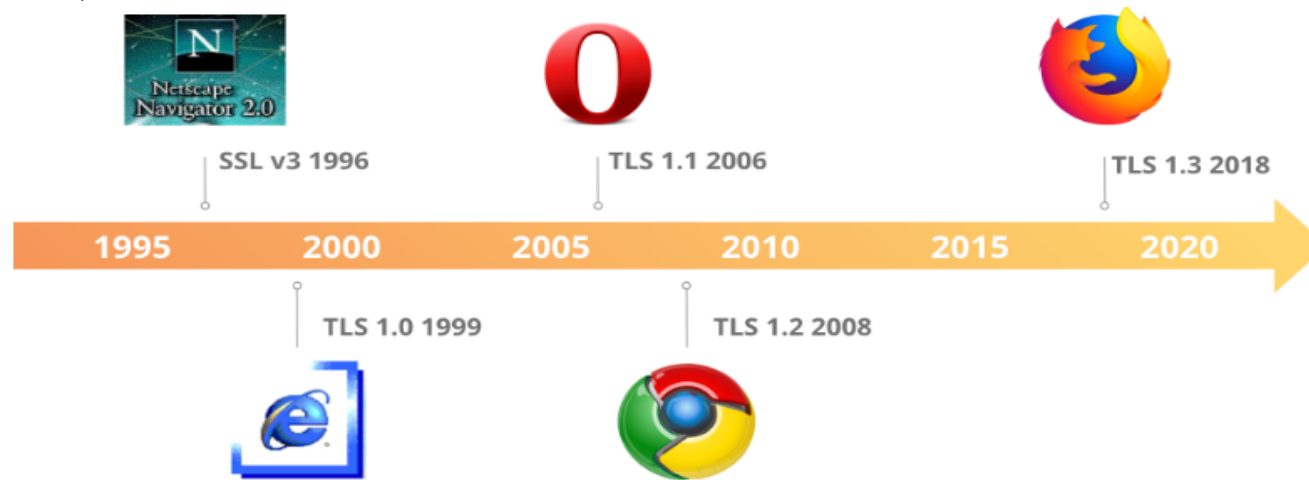
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CS 305 – Computer Networks

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CS 305 – Computer Networks

CSE 5014 – Cryptography and Network Security

Using RSA for Digital Signature

$$S = M^d \bmod n \text{ (RSA signature)}$$

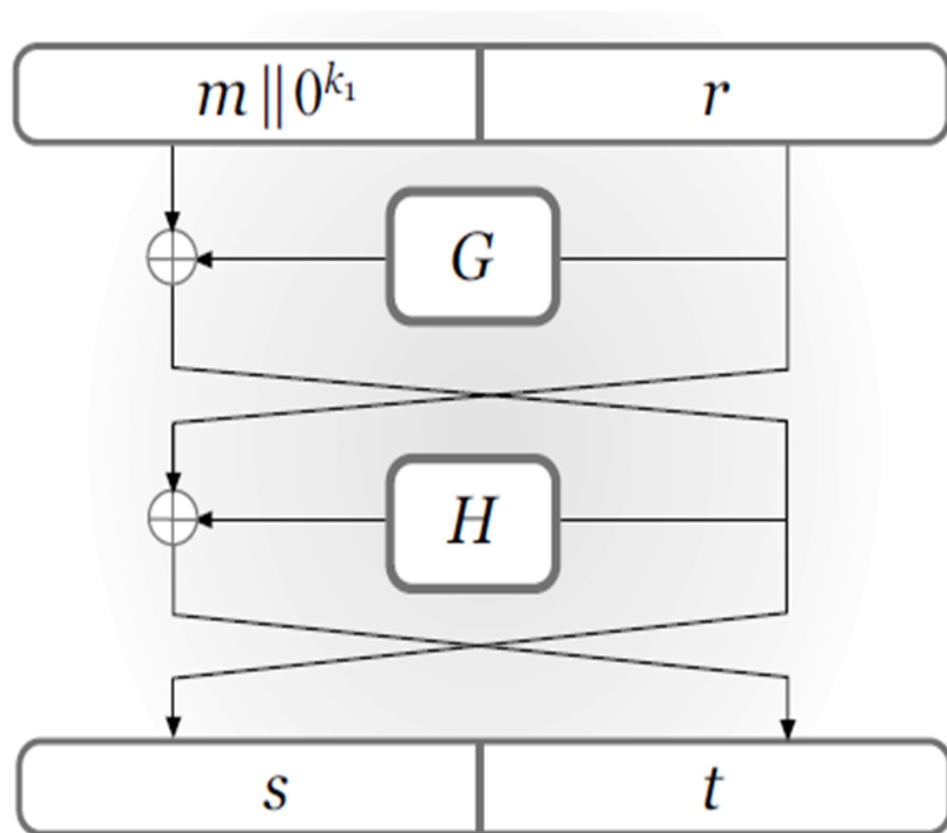
$$M = S^e \bmod n \text{ (RSA verification)}$$

Why?



RSA-OAEP Standard

- RSA-OAEP (Optimal Asymmetric Encryption Padding) is *IND-CCA2 secure*.
- PKCS#1 V2, RFC2437 Standard



The Discrete Logarithm

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Discrete Logarithm Problem:

Given n , b and y , find x .

This is very hard!



El Gamal Encryption

- **Setup** Let p be a prime, and g be a generator of $\mathbb{Z}_p$. The **private key** x is an integer with $1 < x < p - 2$. Let $y = g^x \bmod p$. The **public key** for *El Gamal encryption* is (p, g, y) .

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El Gamal Encryption: Pick a **random** integer k from $\mathbb{Z}_{p-1}$,

$$a = g^k \bmod p$$

$$b = My^k \bmod p$$

The ciphertext C consists of the pair (a, b) .

El Gamal Decryption:

$$M = b(a^x)^{-1} \bmod p$$



Using El Gamal for Digital Signature

$$\begin{aligned}a &= g^k \bmod p \\b &= k^{-1}(M - xa) \bmod (p - 1)\end{aligned}$$

(El Gamal **signature**)

$$y^a a^b \equiv g^M \pmod{p}$$

(El Gamal **verification**)



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Q : How to verify it?



An Example

Choose $p = 2579$, $g = 2$, and $x = 765$. Hence
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► (Public key) $k_e = (p, g, y) = (2579, 2, 949)$

► (Private key) $k_d = x = 765$

Encryption: Let $M = 1299$ and choose a random $k = 853$,

$$\begin{aligned}(a, b) &= (g^k \bmod p, My^k \bmod p) \\ &= (2^{853} \bmod 2579, 1299 \cdot 949^{853} \bmod 2579) \\ &= (435, 2396).\end{aligned}$$

Decryption:

$$M = b(a^x)^{-1} \bmod p = 2396 \times (435^{765})^{-1} \bmod 2579 = 1299.$$



Security of the El Gamal Cryptosystem

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Question 2: Given a ciphertext (a, b) , is it feasible to derive the plaintext M ?

Attack 1: Use $M = by^{-k}$. However, k is **randomly** picked.

Attack 2: Use $M = b(a^x)^{-1} \bmod p$, but x is **secret**.



Diffie-Hellman Key Exchange Protocol

User A

User B

Generate random

$$X_A < p$$

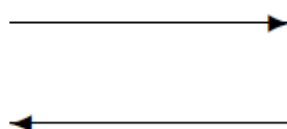
calculate

$$Y_A = \alpha^{X_A} \bmod p$$

Calculate

$$k = (Y_B)^{X_A} \bmod p$$

Y_A



Y_B

Generate random

$$X_B < p$$

Calculate

$$Y_B = \alpha^{X_B} \bmod p$$

Calculate

$$k = (Y_A)^{X_B} \bmod p$$

Cryptography Wonders

- *Digital Signatures*. Electronically sign documents

Zero-knowledge Proofs. Alice proves to Bob that she earns $< \$50k$ without Bob learning her income.

Privacy-perserving data mining. Bob holds DB. Alice gets answer to one query, without Bob knowing what she asked.

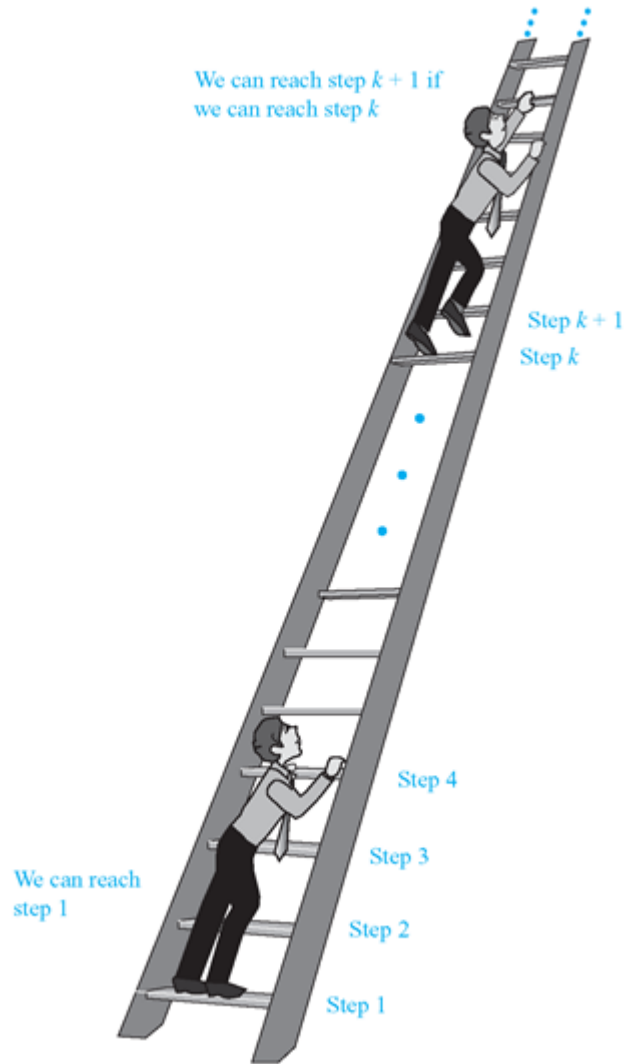
Playing poker over the net. Alice, Bob, Carol and David can play Poker over the net without trusting each other or any central server. (*E-Voting*)

Electronic Auctions. Can run auctions s.t. no one (even not seller) learns anything other than winning party and bid.

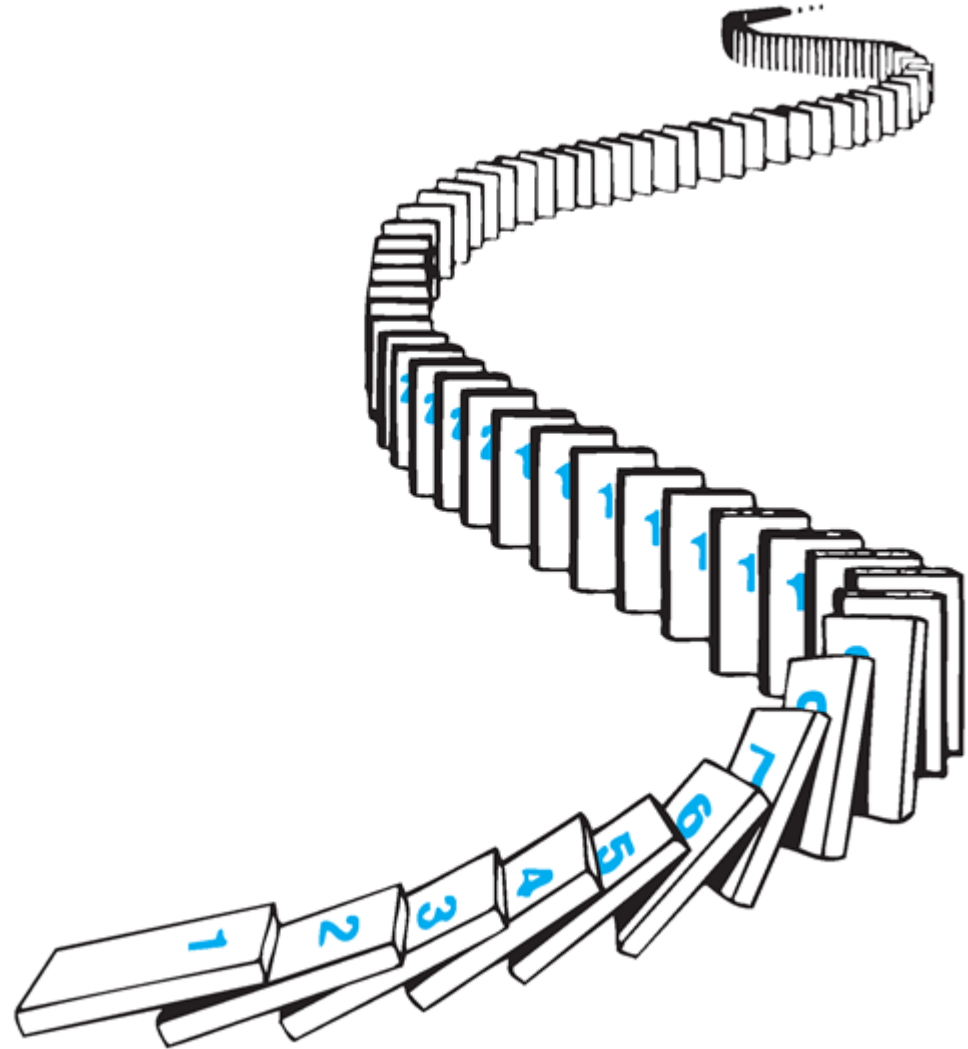
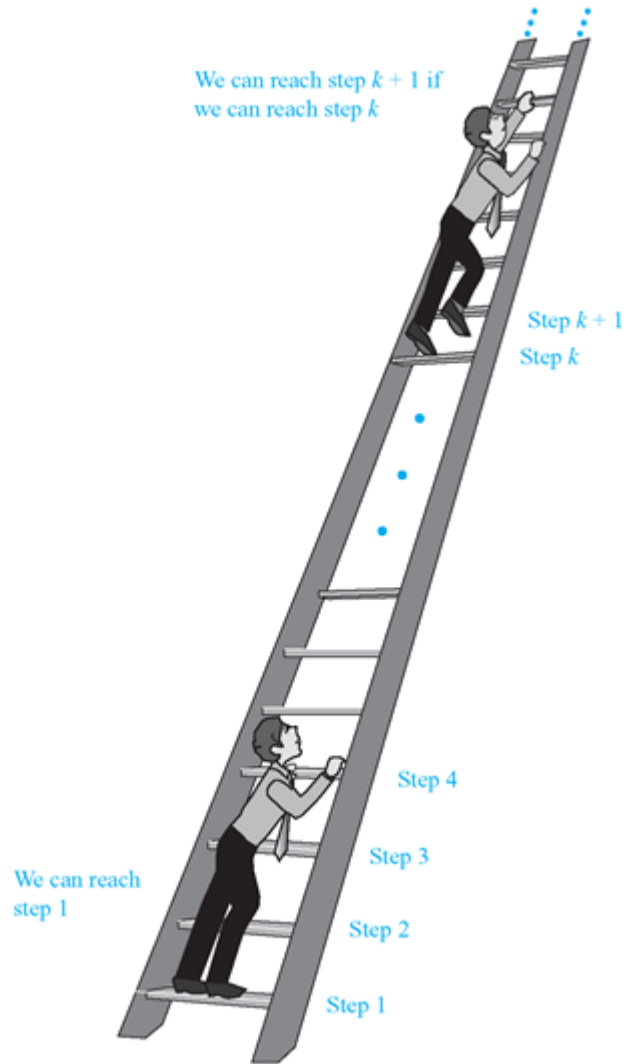
Fully Homomorphic Encryption. Encrypt $E(m)$ in a way that allows to compute $E(f(m))$.



Mathematical Induction



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- We conclude by distinguishing between the *weak principle* of mathematical induction and the *strong principle* of mathematical induction.

The *strong principle* can actually be derived from the *weak principle*.



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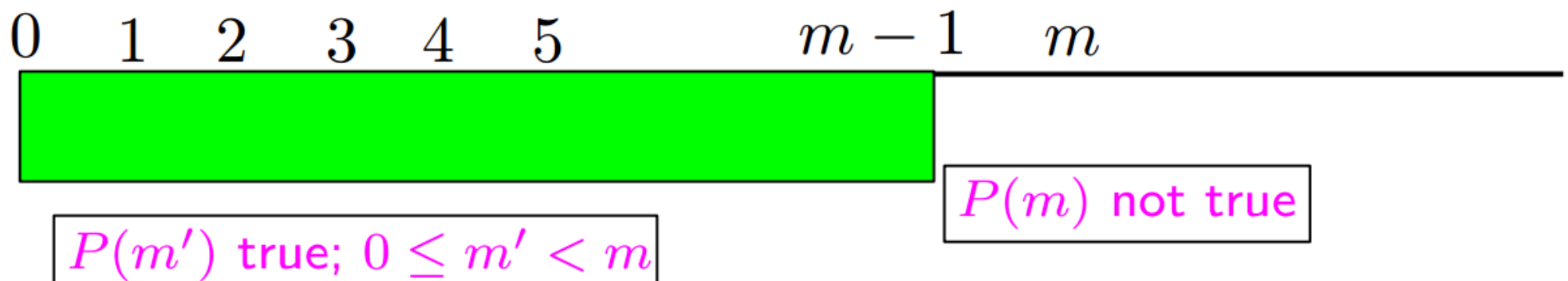


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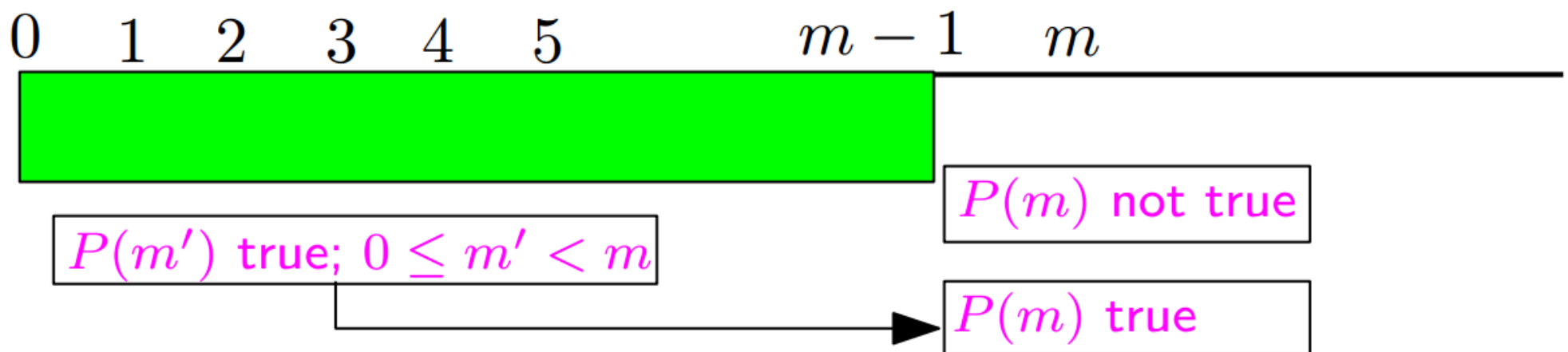


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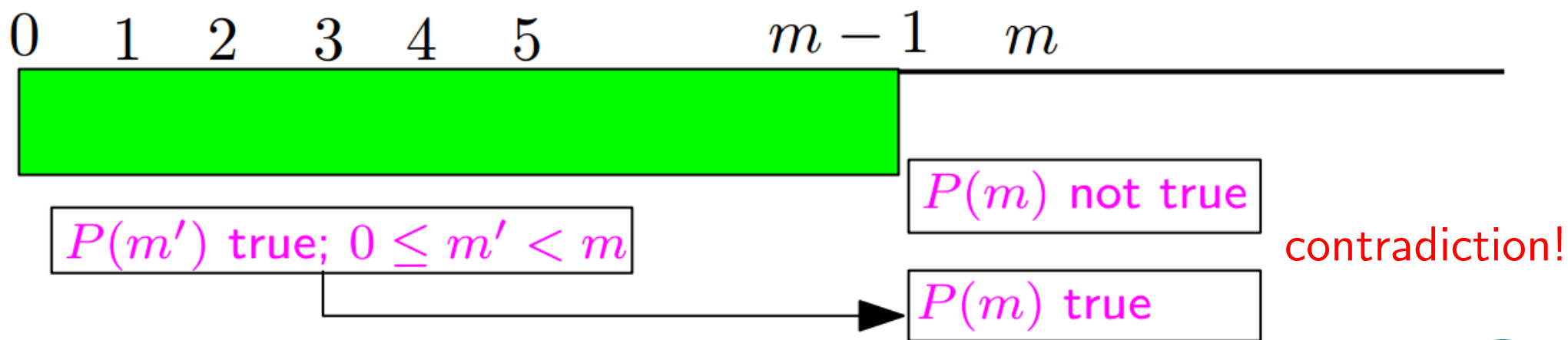


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- ◇ The smallest counterexample n is larger than 0



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◇ Therefore, $(*)$ holds for all positive integers n .



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The **key step** was proving that

$$P(n - 1) \rightarrow P(n)$$

where $P(n)$ is the statement

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When a **for all** quantifier is false, **there must be some n for which it is false**. Let n be the **smallest nonnegative integer** for which $2^{n+1} \not\geq n^2 + 2$.

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This means that, for all $i \in \mathbb{N}$ with $i < n$,

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Then setting $i = n - 1$ gives

$$2^{(n-1)+1} \geq (n-1)^2 + 2.$$

or

$$(*) \quad 2^n \geq n^2 - 2n + 1 + 2 = n^2 - 2n + 3$$



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Thus, we write

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45 - 5



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(a) $P(0)$ is true

(b) if $n > 0$, then $P(n-1) \rightarrow P(n)$



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Since $P(n-1) \rightarrow P(n)$, we see that

$P(0)$ implies $P(1)$, $P(1)$ implies $P(2)$, ...



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- (a) If the statement $P(b)$ is true
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(a) – *Basic Step* *Inductive Hypothesis*

(b) – *Inductive Step* *Inductive Conclusion*



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By mathematical induction, $\forall n > 0, 2^{n+1} \geq n^2 + 2$.



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Proof by Induction

$$\blacksquare \forall n \geq 2, 2^{n+1} \geq n^2 + 3$$

$$\text{Let } P(n) - 2^{n+1} \geq n^2 + 3$$



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By mathematical induction, $\forall n > 2, 2^{n+1} \geq n^2 + 3$.



Proof by Induction

■ $\forall n \geq 2, 2^{n+1} \geq n^2 + 3$

Let $P(n) - 2^{n+1} \geq n^2 + 3$ Base Step

(i) Note that for $n = 2, 2^{2+1} = 8 \geq 7 = 2^2 + 3 - P(2)$

(ii) Suppose that $n > 2$ and that $2^n \geq (n-1)^2 + 3$ (*) Inductive Hypothesis

$$\begin{aligned} 2^{n+1} &\geq 2(n-1)^2 + 6 \\ &= n^2 + 3 + n^2 - 4n + 4 + 1 \\ &= n^2 + 3 + (n-2)^2 + 1 \\ &> n^2 + 3 \end{aligned}$$

Inductive Step

Hence, we've just prove that for $n > 2, P(n-1) \rightarrow P(n)$.

By mathematical induction, $\forall n > 2, 2^{n+1} \geq n^2 + 3$.

Inductive Conclusion



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- ◇ Iterating gives us a proof of $P(n)$ for all n



Strong Induction

- **Principle** (*The Strong Principle of Mathematical Induction*)
 - (a) If the statement $P(b)$ is true
 - (b) for all $n > b$, the statement $P(b) \wedge P(b+1) \wedge \cdots \wedge P(n-1) \rightarrow P(n)$ is true.then $P(n)$ is true for all integers $n \geq b$.

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 - ◇ Then, if n is not a prime power, it is a product of two smaller numbers, each of which is, by the **inductive hypothesis**, a power of a prime or a product of powers of primes.
 - ◇ Thus, by the **strong principle of mathematical induction**, every positive integer is a power of a prime or a product of powers of primes.

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- In reality, they are **equivalent** to each other in that **the weak form is a special case of the strong form, and the strong form can be derived from the weak form.**



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3. We conclude on the basis of the principle of **mathematical induction** that $P(n)$ is true for all $n \geq b$.

Next Lecture

■ recurrence ...

